INCH-POUND

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DETAIL SPECIFICATION

HOOK, HELICOPTER EXTERNAL CARGO, 20,000 POUND CAPACITY, TYPE C

INACTIVE FOR NEW DESIGN AFTER 28 AUGUST 1996

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for a 20,000 pound capacity helicopter external cargo hook, type C.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Air Warfare Center Aircraft Division, Code 414100B120-3, Highway 547, Lakehurst, NJ 08733-5100, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

QQ-C-320	-	Chromium Plating (Electrodeposited)
QQ-P-416	-	Plating, Cadmium (Electrodeposited)

DEPARTMENT OF DEFENSE

MIL-S-4040	-	Solenoid, Electrical, Generals Specification for (Inactive for New Design)
MIL-W-6873	-	Welding; Flash, Carbon and Alloys Steel (Inactive for New Design)
MIL-P-6906	-	Plates, Identification, Aircraft (Inactive for New Design)
MIL-S-7742	-	Screw Threads, Standard, Optimum Selected Series: General Specification for (Inactive for New Design)
MIL-L-7961	-	Lights, Indicators, Press to Test
MIL-A-8625	-	Anodic Coatings for Aluminum and Aluminum Alloys
AFGS-87242	-	Flight Control System General Specification for

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-100	-	Engineering Drawings
MIL-STD-704	-	Aircraft Electric Power Characteristics
MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-889	-	Dissimilar Metals
MS3102	-	Connector, Receptacle, Electric, Box Mounting, Solder Contacts, AN Type (Inactive for New Design)
MS3106	-	Connector, Plug, Electric, Straight, Solder Contacts, AN Type (Inactive for New Design)

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(Unless otherwise indicated, copies of the above specifications and standards are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-B117 - Salt Spray (Fog) Apparatus Operating, Standard Practice for. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE-AMS-STD-2219 - Fusion Welding for Aerospace Applications. (DoD adopted)
SAE-AMS-W-6858 - Welding, Resistance: Spot and Seam. (DoD adopted)

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. Materials used in the manufacture of the cargo hook shall meet the requirements of this specification.

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3.2.1 Metals. All metals used in the construction of the cargo hook shall be of the corrosion resistant type or shall be treated to resist corrosion in accordance with MIL-A-8625, QQ-C-320 or QQ-P-416. Dissimilar metals shall not be used in accordance with MIL-STD-889.

3.3 Design and construction. The hook shall be automatic (see 6.3.1) or semiautomatic (see 6.3.3) cargo pickup.

3.4 Screw threads. All screw threads shall be in accordance with MIL-S-7742.

3.5 Threaded parts. All threaded parts shall use aircraft locking methods. Lock washers shall not be used for locking.

3.6 Welding. Welding, where used, shall be in accordance with SAE-AMS-6858 and SAE-AMS-STD-2219.

3.7 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Change in manufacturer's part numbers shall be in accordance with MIL-STD-100.

3.8 Operating condition. The cargo hook shall operate under the following environmental conditions:

3.8.1 Altitude. The hook shall operate at any altitude between sea level to 20,000 feet (see 4.5.2.14).

3.8.2 Temperature. The hook shall operate at any ambient temperature between –65 °F to +160 °F (see 4.5.2.10 and 4.5.2.11).

3.8.3 Vibration. The hook shall withstand the helicopter vibrational environment (see 4.5.2.9).

3.8.4 Sand. The hook shall withstand the blowing sand environment (see 4.5.2.13).

3.8.5 Salt spray. The hook shall withstand the salt spray environment (see 4.5.2.12).

3.8.6 Shock. The hook shall withstand the functional shock environment (see 4.5.2.17).

3.8.7 Water immersion. The hook shall withstand salt water immersion without permanent damage or operational impairment (see 4.5.2.15).

3.8.8 Fungus. The hook shall withstand the fungus environment (see 4.5.2.16).

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3.9 Detailed requirements.

3.9.1 General. The hook shall be of the open-throat type. The load sling adapter ring or other pickup device may be introduced into the load bearing portion of the hook automatically or in any event it shall not require personal contact with the hook. There shall be no protuberances or other obstacles to prevent or hinder engagement with the load sling adapter ring or other pickup device. The open-throat shall face forward when installed on an aircraft.

3.9.1.1 Throat. The throat (see 6.3.5) configuration shall facilitate engagement with the load sling adapter or other pickup device. The throat shall accept any of the following sling adapter rings or pickup devices or their equivalents:

a. Nylon strap ring 2.75-inch wide and 0.875-inch thick.

b. A steel ring made of 1.5-inch diameter stock.

c. Two 1.5-inch diameter ropes.

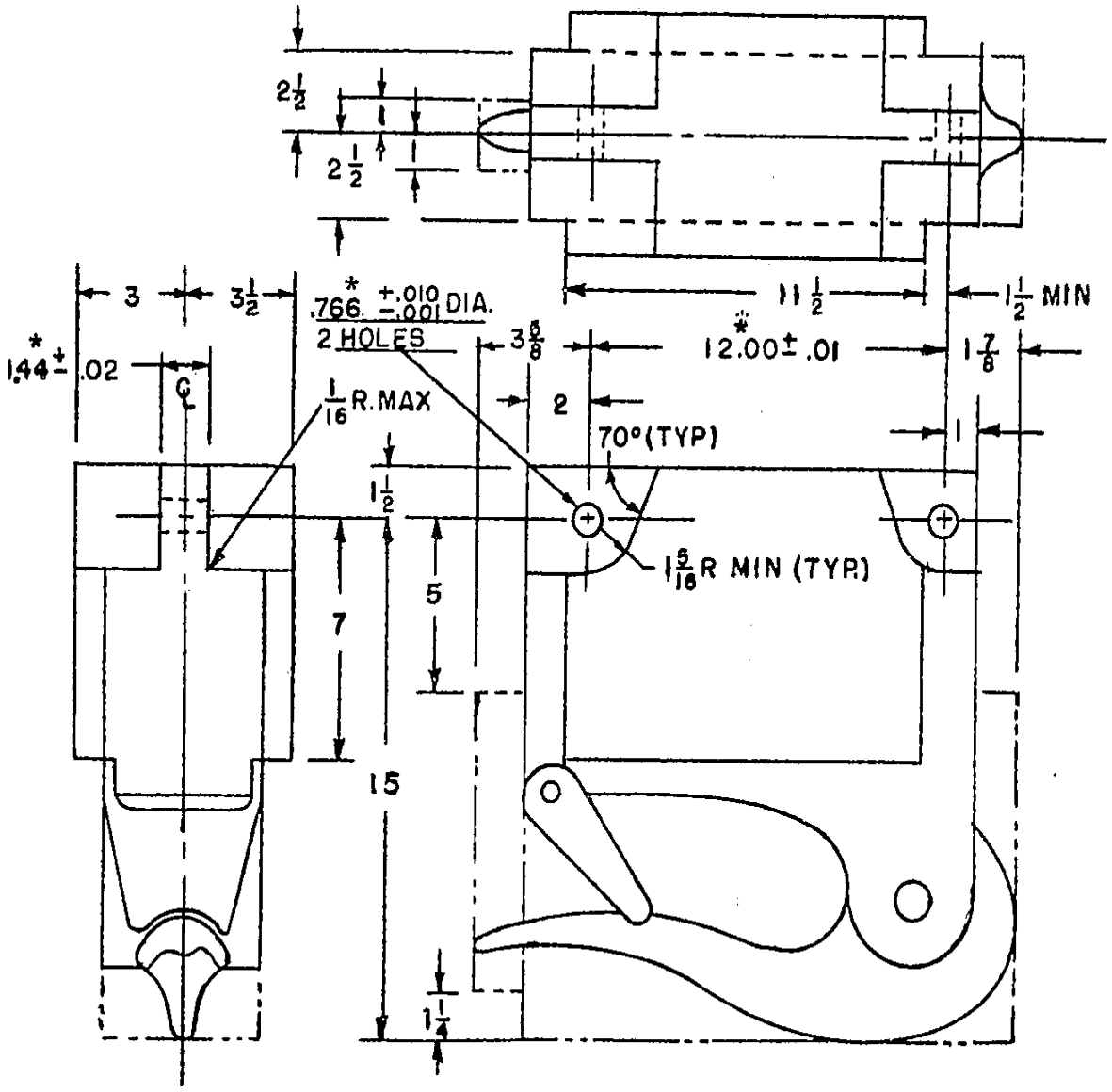
3.9.1.2 Housing. The hook housing shall be constructed to support and enclose the components and to protect them from environmental conditions. Components subject to water damage shall be enclosed within a completely waterproof housing or other protective device.

3.9.1.3 Load beam. The load beam shall support all loads under all operational conditions. The load beam shall be mounted in the hook housing by a pivot near its aft end such that upon release the front end will swing downward and aft. The configuration of the load beam shall assist the load sling adapter ring to position itself in the hook while presenting minimum resistance to disengagement on release.

3.9.1.4 Keeper. A keeper shall be used in the hook throat to positively retain the load sling adapter ring in the desired position in the hook under all operational flight conditions until release. The keeper shall permit engagement of pickup devices consisting of more than one part or ring, each engaged separately without personal contact with any part of the hook.

3.9.1.5 Attachment. The hook shall have two attachment holes on its upper portion; one near the forward end and one near the aft end as shown on figure 1, such that the point at which the cargo load is applied is midway between them. The suspension holes shall attach to a semi-rigid mounting structure (see 6.3.4) beneath the aircraft and shall permit removal of the hook when desired.

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NOTES: 1. All dimensions marked with an asterisk (*) must be held.
2. All other dimensions define maximum envelope size except as noted.
3. All dimensions are in inches.

FIGURE 1. Envelope dimensions.

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3.9.1.6 Release mechanism. The internal release mechanism in the hook shall provide positive retention of the load beam in its load carrying position until the release mechanism is actuated by one of the modes of release specified in 3.9.6.1. Upon actuation, the release mechanism shall react so as to remove all restraint from the load beam. The load beam shall then open and completely release the load. Components of the release mechanism which rely on a wiping or rolling action between members shall be self-cleaning to prevent contamination and fouling.

3.9.2 Weight. The maximum weight of the hook shall be not greater than 40 pounds.

3.9.3 Envelope. The overall hook dimensions shall be not greater than those shown on figure 1.

3.9.4 Capacity. The hook shall have a 20,000-pound design 1oad rating. The hook shall operate under loads specified herein, at attitudes from horizontal to a 30-degree nose pitch-up. Application of a torsional moment to the hook while under rated load, as specified herein, shall not damage the hook.

3.9.5 Design load factors. The hook shall withstand a design limit load (static) of 60,000 pounds with no yield or damage to working parts, and a design ultimate load (static) of 90,000 pounds.

3.9.6 Operating features.

3.9.6.1 Hook release. The hook shall release the cargo by the following methods. Actuation of any of the releases for a continuous period of 30 seconds shall not damage the hook or its components.

3.9.6.1.1 Normal release. The normal release shall be actuated when the required aircraft voltage is applied. The hook release device shall release all loads between seven to 6,000 pounds in not greater than 0.5-second from initiation to release. The device may be an electrical solenoid in accordance with MIL-S-4040 or an electrical actuator in accordance with AFGS-87262.

3.9.6.1.2 Automatic touchdown release. Automatic touchdown release shall be actuated when the required aircraft voltage is applied and the load is reduced to 150 ± 25 pounds by ground contact. The hook shall release all loads greater than seven pounds. The hook shall remain in the released condition until the circuit is deactivated. The normal release mode shall not be deactivated when the automatic touchdown mode is used.

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3.9.6.1.3 Manual release. The manual release shall be actuated when a link or cable from the helicopter to the hook is pulled. The release shall release all loads from seven pounds to the 60,000-pound design limit. The hook need not be operable after release of loads between the rated capacity and the design limit load but shall return to full operating condition with only minor repairs or replacement of minor parts. The hook and its components shall incur no damage from releases of loads up to rated capacity in this mode. The stroke through which the actuating linkage must travel to execute a complete release of the hook shall be not less than 0.5-inch nor greater than 1.25 inches including any pretravel. The amount of pretravel shall be not greater than the actuating travel. The point at which the actuating force is applied to the mechanism shall be so located that the mechanism shall release when a connection is made between it and a vertically arising point located four inches above the mounting holes, two inches to starboard of the fore-and-aft hook centerline, and not more than two inches forward or aft of the midpoint between the mounting holes. The configuration of the manual release attachment point shall accommodate a standard AN or MS terminal at the end of a 0.0625-inch diameter cable. The force required to actuate this release under various load conditions shall be not greater than the following:

LOAD	FORCE
None	15 lbs
20,000 lbs	50 lbs
60,000 lbs	80 lbs

The release mechanism shall provide a force of 9 ± 1 pounds at the point of attachment of the actuating linkage in a direction which will retain the aircraft linkage in its de-actuated position. This force is included in the release forces specified above.

3.9.6.1.4 Ground release. The ground release shall actuate when a link or cable from ground personnel to the hook is pulled. The force required to actuate the release mechanism of the hook by this method shall be not greater than 15 pounds under no-load condition. The release mechanism shall not be actuated by falling or stringing objects, such as cables or heavy weights.

3.9.6.2 Relatch. After release of any load up to rated capacity, the hook load beam shall automatically relatch into its load carrying condition within five seconds after the release controls have returned to the non-release position. In the event the load beam does not relatch after release of a load, reactuation of either the normal or manual release shall cause the load beam to relatch.

3.9.6.3 Indication of unlatched load beam. There shall be a means on the hook to indicate when the load beam is in an unlatched condition. The device shall complete the circuit when this condition occurs, illuminating an aircraft light in accordance with MIL-L-7961.

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3.9.7 Electrical characteristics.

3.9.7.1 Electric power. The hook shall operate with electric power in accordance with MIL-STD-704, table II, at 28 volts direct current (vdc), 15 amperes.

3.9.7.2 Low input voltage. The hook shall accomplish a normal release with an input of 22 vdc.

3.9.8 Electrical connections. An electrical receptacle, MS3102El4S-5P, shall be provided on the hook for interconnection of the hook electrical circuits with those in the helicopter. The following pin schedule shall be used:

PIN DESIGNATION	FUNCTION
A	Automatic Touchdown Arming
B	Normal Release
C	Ground
D	Load Beam Unlatched Indication
E	Spare

The receptacle shall be positioned to permit entry of the mating MS3106E14-5S plug from one of the following directions; top or forward end, without the plug and related hardware obstructing the mounting provisions or projecting outside the dimensional envelope shown on figure 1.

3.10 Lubrication. The hook and all components shall operate with the lubricant for the maintenance period specified in 3.11.2. Lubrication (see 6.4) shall not contribute to fouling due to entrapment of foreign matter.

3.11 Maintenance.

3.11.1 Time required for maintenance. Time required for maintenance including disassembly, inspection, cleaning, and reassembly of the hook shall be not greater than three man-hours.

3.11.2 Maintenance inspection period. The hook shall operate with the following maintenance inspection period:

a. A sight inspection and operational check in the normal, manual, and automatic release conditions every 60 hours of aircraft time, or 60 days, whichever occurs first. Hook loadings shall provide a reliable indication of proper operation.

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b. Inspection, lubrication if required, and full-load operational check in all operating modes every 240 hours of aircraft time, or six months, whichever occurs first.

c. Complete disassembly and overhaul every 5,000 cycles of hook operation, or two years, whichever occurs first.

3.12 Nameplate. A nameplate, giving the following information, shall be provided and securely attached to the hook. The nameplate shall be in accordance with MIL-P-6906.

Hook, Helicopter External Cargo, type C
Rated Capacity: 20,000 pounds
Stock Number:_____
Name of Manufacturer:_____
Manufacturer's Part No.:_____
Manufacturer's Serial No.:_____
Contract No.:_____
Specification No. MIL-DTL-81014 U.S. Property

3.13 Instruction plate. An instruction plate shall be attached to the hook depicting the schematic wiring diagram of the hook.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. One test sample shall be tested. The test sample shall be identical to the production hooks and shall be identified with the manufacturer's own part number. The test sample shall be subjected to and pass all the tests specified in 4.5.

4.3 Conformance inspection. Each cargo hook submitted shall be subjected to examination (see 4.5.1) and the operational load test (see 4.5.2.1).

4.4 Test conditions Unless otherwise specified, all inspections shall be performed at prevailing ambient conditions with electric power specified in 3.9.7.1. The hook shall be rigidly or semi-rigidly mounted, as applicable, in a test structure. The load shall be applied with a nylon sling not less than four feet long. The order of testing shall be as from 4.5.2.2 to 4.5.2.17.

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4.5 Tests.

4.5.1 Examination of product. The cargo hook, its subassemblies and components shall be examined to determine compliance to this specification with respect to materials and markings.

4.5.2 Test procedures.

4.5.2.1 Operational load test. The operational load test shall consist of a specified number of operational cycles. These cycles shall consist of:

a. Engagement of load.

b. Hold load for two minutes.

c. Release load with releases indicated below.

d. Relatch automatically.

The following loads and release methods shall be used for the indicated number of cycles:

NO. OF CYCLES	LOAD	RELEASE METHOD
10	6,000 lbs.	Normal
15	20,000 lbs.	Automatic Touchdown
5	7 lbs.	Normal
5	7 lbs.	Ground
2	20,000 lbs.	Manual
5	5,000 lbs.	Manual

After half of each of the above tests have been completed, the forces required to actuate the manual release shall be checked for no-load and full-load conditions. The release forces shall not exceed the values specified in 3.9.6.1.3.

4.5.2.2 Overload test. A static test load of 60,000 pounds shall be suspended from the hook for one minute. The hook shall suffer no yield or damage and shall operate through two cycles of each of the tests specified in 4.5.2.1.

4.5.2.3 Overload release test. The cargo look manual release shall be actuated one time with a 60,000 pound load suspended from the hook. The hook shall then be examined to see that there is no major damage and to ascertain that the hook can be returned to operating condition with only minor repair and replacement of minor parts. After examination and return of the hook to its normal operating condition, the hook shall operate.

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4.5.2.4 Torque test. A load equal in magnitude to the rated operating capacity of the hook shall be suspended from the hook using a six-inch inside diameter steel ring, made from stock, to simulate a load pickup adapter ring. A torque of 10,000 inch-pounds shall then be applied to the ring for two minutes. Upon completion, the direction of the torque shall be reversed, and the procedure repeated. The hook shall suffer no permanent damage and shall operate through one release of each of the tests specified in 4.5.2.1.

4.5.2.5 Angular load test. The cargo hook shall be rigidly suspended from a rigid structure and pitched up such that there is an angle of 30 degrees between the longitudinal axis of the hook and the horizontal. The hook shall then have vertical loads applied, gradually increasing from zero to rated capacity and returning to zero. The loading cycle shall be not less than two minutes. The hook shall be subjected to 30 cycles. Upon completion of the test, the hook shall be inspected for permanent damage and shall be operated at this pitched attitude through one cycle of each of the tests specified in 4.5.2.1.

4.5.2.6 Low voltage test. A load of 6,000 pounds shall be suspended from the hook and the normal release circuit energized with an input of 22 ± 0.5 vdc. The hook shall release the load and relatch automatically.

4.5.2.7 Maximum current drain. An ammeter shall be installed in the input power supply. The hook shall be operated through three normal releases while being subjected to a 6,000-pound load with an input of 28 ± 0.5 vdc. The maximum current drain shall be not greater than 15 amperes.

4.5.2.8 Endurance test. The endurance test shall consist of the hook passing two complete series of cyclic tests, performed in accordance with the cycles below. Overhaul of the hook is permissible between the two series of tests. Under the overload (three times rated load) condition, replacement of damaged minor parts is permissible in accordance with 4.5.2.3.

CYCLES	RELEASE MODE	LOAD
50	Normal	7 lbs.
1,500	Automatic	10,000 lbs.
1	Manual	7 lbs.
500	Manual	1,000 lbs.
250	Manual	5,000 lbs.
1,500	Normal	3,000 lbs.
750	Automatic	20,000 lbs.
750	Normal	6,000 lbs.
50	Manual	10,000 lbs.
25	Manual	20,000 lbs.
1	Manual	60,000 lbs.

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Measurements shall be taken before and after the endurance test and a wear analysis shall be made showing the wear on each rubbing part. If during that portion of the test using the normal, or automatic touchdown mode, the hook fails to release, the manual release shall be actuated. If the hook then fails to release it shall be considered as an additional release failure. The number of release failures during the endurance tests shall be not greater than two, not more than one of which shall be in the manual mode. During the endurance tests, the hook shall be allowed to relatch automatically. Not more than ten relatch failures shall be permitted during the complete endurance test. Failure to relatch at loads in excess of the rated capacity of the hook shall not constitute a failure. The forces required to actuate the manual release shall be checked for no-load, full-load, and 60,000 pounds after each of the series of tests constituting the endurance tests. The forces shall be not greater than the values specified in 3.9.6.1.3 (see 6.5).

4.5.2.9 Vibration test. The hook shall be subjected to a vibration test in accordance with MIL-STD-810, Method 514.5, Procedure I, Category 18. Upon completion of this test the hook shall be rigidly attached to a supporting fixture and the hook loaded. A vibrator producing resonant frequencies in the hook under all loads stated below, shall be attached to the fixture or the hook. All tests shall be conducted at 2.5 ± 0.5g acceleration level at the hook. The hook shall be vibrated for the times and under the loads noted along each of its three mutually perpendicular axes. The noted times are totals and shall be equally divided among testing on each of the three axes. The tests shall consist of cycling vibration and resonant vibration with test time divided equally between them. The cycling vibration shall be conducted between 20 and 300 cycles per second and resonant frequencies in the hook shall be noted. Resonant vibration shall be performed at all of the major resonant points with equal time being used for each point when more than one resonant point is found. The hook release shall be actuated while the hook is at a resonant point at the intervals and under the loads noted below. No inadvertent releases shall be permitted.

LOAD	TOTAL TIME	INTERVAL OF RELEASE	MODE OF RELEASE
7 lbs.	0.5 hour	Once on each axis of vibration	Normal
6,000 lbs.	6 hours	Three times on each axis of vibration *	Normal
20,000 lbs.	4 hours	One time on each axis of vibration **	Manual

* At least one release shall be made for each resonant point, up to three if more than one resonant point is found.

** While at the major resonant point.

4.5.2.10 High temperature test. The hook shall be tested in accordance with MIL-STD-810, Method 501.4, Procedure II, table 501.4-I, induced conditions, for 48 hours. The hook shall be tested through one cycle of each of the tests specified in 4.5.2.1.

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4.5.2.11 Low temperature test. The hook shall be tested in accordance with MIL-STD-810, Method 502.4, Procedure II. The hook temperature shall be stabilized at –65 °F. The hook shall be tested through one cycle of each of the tests specified in 4.5.2.1.

4.5.2.12 Salt spray test. The hook shall be tested in accordance with ASTM-B117 for a period of 48 hours. The hook shall be tested through one cycle of each of the tests specified in 4.5.2.1 without removal of any salt deposits resulting from the test.

4.5.2.13 Sand test. The hook shall be tested in accordance with MIL-STD-810, Method 510.4, Procedure II, except that the sand velocity through the test chamber shall be maintained at 2,500 ± 500 feet per minute through the first half of the test and between 100 and 500 feet per minute through the second half of the test. Upon completion of the test, the hook shall be inspected for deterioration and shall be operated through one cycle of each the tests specified in 4.5.2.1.

4.5.2.14 Altitude test. The hook shall be tested in accordance with MIL-STD-810, Method 500.4, Procedure I, at an altitude of 20,000 feet and an ambient temperature of –65 °F. The hook shall be operated through one cycle of each test condition specified in 4.5.2.1 after the test.

4.5.2.15 Water immersion tests. The hook shall be tested by immersion in sea water or water of equivalent salinity, to a depth of five feet for 15 minutes, and after being withdrawn the hook shall be operated through one cycle of each test condition specified in 4.5.2.1. A fresh water washdown is permissible after completion of operations.

4.5.2.16 Fungus test. The hook shall be tested in accordance with MIL-STD-810, Method 508.5 for 28 days. Upon completion of this test, the hook shall be operated through one cycle of each test condition specified in 4.5.2.1.

4.5.2.17 Shock test. The hook shall be tested in accordance with MIL-STD-810, Method 516.5, Procedure I, functional test for flight equipment. Three shocks in each direction shall be made for a total of 18 shocks. Upon completion of this test, a load shall be attached and dropped so as to induce a shock loading equivalent to 20,000 pounds at 3 g's. The hook shall successfully sustain ten such shocks. Upon completion of this test the hook shall be operated through one cycle of each test condition specified in 4.5.2.1.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military

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Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The cargo hook is intended to provide a helicopter with an external means for cargo pickup, transport, and release when installed and connected to the controls within the helicopter.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DoDISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).

c. When first article is required (see 3.1).

d. Packaging requirements (see 5.1).

6.3 Definitions.

6.3.1 Automatic cargo pickup. An automatic cargo pickup is when the helicopter pilot flies the aircraft so that the hook engages with the load without requiring the aid of ground personnel to physically engage the hook with the load.

6.3.2 Rigid mount. A rigid mount is when the cargo hook is attached to the supporting structure so that there is no relative motion between them.

6.3.3 Semiautomatic pickup. A semiautomatic pickup require is ground personnel to engage the load sling in the hook but without having contact with the hook.

6.3.4 Semi-rigid mount. A semi-rigid mount is when the cargo hook is attached to a member of the supporting structure such that the hook and member can pivot or swing about a longitudinal axis located just above the hook.

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6.3.5 Throat. The throat is that area between the hook housing and the load beam, open at the forward end, and culminating in the area where the load beam supports the sling adapter ring while transporting a cargo load.

6.4 Lubrication. For guidance on lubrication see MIL-HDBK-838.

6.5 Endurance test failures. After any release failure in the manual mode, and subsequent to actuation of the manual release after a release failure in the normal mode, the hook may be examined to determine the cause of the failure. If the fault is determined for either a release or relatch failure, the deficiency should be corrected. If the fault is not determined, the tests may be continued. If the fault can be corrected by cleaning, adjustment, lubrication, or minor dressing of surfaces, the hook should be reassembled and the testing continued. If corrective action requires general overhaul or replacement of parts other than common fasteners, the testing should be started from the beginning. Should more than two release failures, or more than ten relatch failures occur during the endurance tests, the cause of failures should be determined, corrective action should be taken, and the testing started from the beginning.

6.6 Subject term (key word) listing.

Beam, load
Housing
Keeper
Release mechanism
Suspension
Throat

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS

Preparing activity:
Navy - AS

(Project 1670-0003)

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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-DTL-81014B	**2. DOCUMENT DATE** (*YYYYMMDD*) 20010202

3. DOCUMENT TITLE
HOOK, HELICOPTER EXTERNAL CARGO, 20,000 POUND CAPACITY, TYPE C

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

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